# EE 474 Review Exam 1

Name _____

- Answer each of the questions. Show your work.
- Note were essay-type answers are requested. Answer with complete sentences. Incomplete sentences will count heavily against the grade.
- Write your answers on this handout, attach extra sheets as necessary. Put your name on any extra sheets.
- This test is closed-book. Only one 8.5x11 page (double sided) of notes may be used. No other written materials are allowed.
- A non-programmable calculator is allowed. No other electronic device is allowed. Put away all other electronic devices and turn off cell phones.
- **A note on calculations**: if calculations are required, it is sufficient to write the equation of the calculation that gives the term. For example:

  If the cosine of $a$ is required, it is sufficient to write $\cos(a)$ .

- There is an extra credit problem you should consider if you have extra time.

**Do not execute any program on a programmable calculator.**

Problem 1 _____ (40 points)

Problem 2 _____ (20 points)

Problem 3 _____ (20 points)

Problem 4 _____ (20 points)

Extra Credit _____ (15 points)

## Notes (information that may be helpful for the exam)

| Signal | Description | Units | Signal | Description | Units |
|---|---|---|---|---|---|
| $v(t)$ | voltage | [volts] | $i(t)$ | current | [amps] |
| $x(t)$ | position | [meters] | $F(t)$ | force | [Newtons] |
| $\dot{x}(t) = v(t)$ | velocity | [m/sec] | | | |
| $\theta(t)$ | angle | [radians] | $T(t)$ or $M(t)$ | torque (or moment) | [Newton-meters] |
| $\dot{\theta}(t) = \omega(t)$ | angular velocity | [rad/sec] | | | |

Table 1: Some signals used to analyze motion control system.

| Linear motion | | | Rotational motion | | |
|---|---|---|---|---|---|
| Term | Modeling equation | Units | Term | Modeling equation | Units |
| Mass | $m\ddot{x}(t) = f_{net}(t)$ | $[Kg] = \left[\frac{N}{m/s^2}\right] = \left[\frac{N-s^2}{m}\right]$ | Inertia | $I\ddot{\theta}(t) = M_{net}(t)$ | $\left[Kg-m^2\right] = \left[\frac{N-m}{rad/sec^2}\right] = \left[\frac{N-m-s^2}{rad}\right]$ |
| Spring | $f_k(t) = k(x_1(t) - x_2(t))$ | $\left[\frac{N}{m}\right]$ | Rotational spring | $M_k(t) = k(\theta_1(t) - \theta_2(t))$ | $\left[\frac{N-m}{rad}\right]$ |
| Damper | $f_b(t) = b(\dot{x}_1(t) - \dot{x}_2(t))$ | $\left[\frac{N}{m/s}\right] = \left[\frac{N-s}{m}\right]$ | Rotational damper | $M_b(t) = b(\dot{\theta}_1(t) - \dot{\theta}_2(t))$ | $\left[\frac{N-m}{rad/sec}\right] = \left[\frac{N-m-s}{rad}\right]$ |

Table 2: Basic parameters of mechanical systems.

- Example transfer function in rational polynomial form:

$$H(s) = \frac{b_1 s + b_0}{a_2 s^2 + a_1 s + a_0} \tag{1}$$

- Example transfer function with monic numerator and denominator polynomials, and $K_{rlg}$

$$H(s) = K_{rlg} \frac{s + b_0}{s^2 + a_1 s + a_0} \tag{2}$$

- Quadratic equation, given the polynomial

$$as^2 + bs + c = 0$$

the roots are:

$$s = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \tag{3}$$

- Graphical residue method:

$$C_k = K_{rlg} \frac{\text{Product of vectors from zeros}}{\text{Product of vectors from poles}}$$

## Problem 1.

Model the mechanism of figure 1.
(Using the 5 steps presented in lecture).

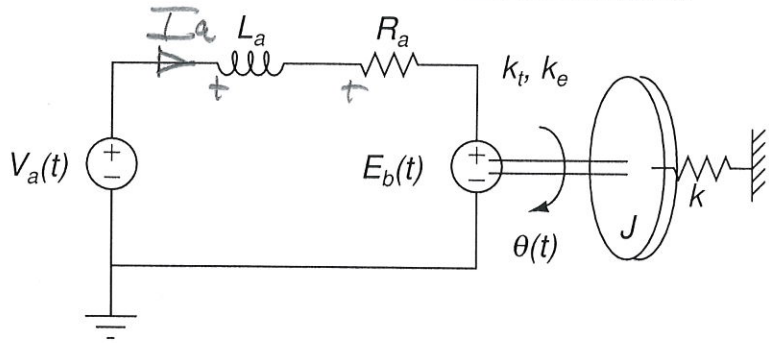



Figure 1: DC motor drive. The input is applied voltage $V_a(t)$. The output is position $\theta(t)$.

1. Setup the model

   (a) Make lists of signals and parameters, including symbol and units (all units in meters, kilograms, seconds)

   (b) Write the constituent equations for the elements of the system

Signals

| | | |
|---|---|---|
| $V_a(t)$ | App. Volts | Volts |
| $I_a(t)$ | Current | Amps |
| $E_b(t)$ | BACK EMF | Volts |
| $\theta(t)$ | Rotation | RAD |

Params

| | | |
|---|---|---|
| $L_A$ | Ind. | Henries |
| $R_A$ | Res. | Ohms |
| $K_E$ | Motor BEMF | $\frac{V}{RAD/S}$ |

| | | |
|---|---|---|
| J | Inertia | $\frac{N\text{-}m}{RAD/S^2}$ |
| K | Spring | $\frac{N\text{-}m}{RAD}$ |
| $K_t$ | Motor Torque | $\frac{N\text{-}m}{Amp}$ |

Electrical

$$V_{LA} = L_a \frac{d}{dt} I_a$$

$$V_{RA} = R_a I_a$$

$$E_b(t) = K_e \dot{\theta}(t)$$

Mechanical

$$J\ddot{\theta}(t) = T_{net}(t)$$

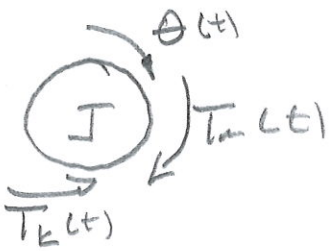
$$T_m(t) = K_t I_a(t) \quad \text{Motor Torque}$$

$$T_k(t) = k\,\theta(t) \quad \text{Spring}$$

2. Develop equations for the system,

   (a) Draw the free body diagram (for mechanical system).

   (b) Write the continuity equation describing the connections of the system.




Mech: $T_{net}(t) = T_m(t) - T_{ka}(t)$

Elect: $V_a(t) = V_{La}(t) + V_{Ra}(t) + E_b(t)$

3. Put the governing differential equations in standard form (output signal on left, input signals on right).

Mechanical: $J\ddot{\theta}(t) = T_m(t) - T_L(t)$

$= K_t I_a(t) - K\theta(t)$

gives $J\ddot{\theta}(t) + K\theta(t) = K_t I_a(t)$

Electrical $L\frac{d}{dt} I_a(t) + R I_a(t) = V_a(t) - K_e \dot{\theta}(t)$

4. Write the transfer function, $G_{uy}(s) = \theta(s)/V_a(s)$.

- Convert all diff. eq's to the frequency domain.
- Algebraically derive $G_{uy}(s)$. (Use the back of the previous sheet for more space).
- Finally, put $G_{uy}(s)$ in rational polynomial form.

In Freq Domain

$(Js^2 + K)\theta(s) = K_t I_a(s)$ (1)

$(L_a s + R_a) I_a(s) = V_a(s) - K_e s\theta(s)$ (2)

Solving (1) for $I_a$ & plugging into (2) gives

$I_a(s) = \left(\frac{Js^2 + K}{K_t}\right)\theta(s)$

$(L_a s + R_a)\left(\frac{Js^2 + K}{K_t}\right)\theta(s) = V_A(s) - K_e s\theta(s)$

Move all terms in $\theta(s)$ to left

$\left((L_a s + R_a)(Js^2 + K) + K_t K_e s\right)\theta(s) = K_t V_A(s)$

$s^3$ $s^2$ $s^1$ $s^0$

$L_A J$ $R_a J$ $L_A K$ $R_a K$

$K_t K_e$

$$\frac{\theta(s)}{V_a(s)} = \frac{K_t}{L_A J s^3 + R_a J s^2 + (L_A K + K_t K_e)s + R_a K}$$

5. Validate that units balance in the transfer function. (Keep in mind that radians are dimensionless).

Repeat

$$\frac{\Theta(s)}{V_a(s)} = \frac{K_t}{L_A J s^3 + R_A J s^2 + (L_A f + K_t K_e) s + R_A f}$$

Gives $\frac{N\text{-}m}{Amp}$

$$\frac{\frac{V\text{-}s}{A}\ \frac{N\text{-}m\text{-}s^2}{rad}\ s^{-3} + \frac{V}{A}\ \frac{N\text{-}m\text{-}s^2}{rad}\ s^{-2} + \left(\frac{V\text{-}s}{A}\ \frac{N\text{-}m}{rad} + \frac{N\text{-}m}{A}\ \frac{V\text{-}s}{rad}\right) s^{-1} + \frac{V}{A}\ \frac{N\text{-}m}{rad}}{-}$$

All terms in Denominator have units

$\frac{V}{A}\ \frac{N\text{-}m}{rad}$

$$\frac{\Theta}{V_a}: \frac{\frac{N\text{-}m}{A}}{\frac{V}{A}\ \frac{N\text{-}m}{rad}} = \frac{rad}{volt} \quad \checkmark$$

**Problem 2.**

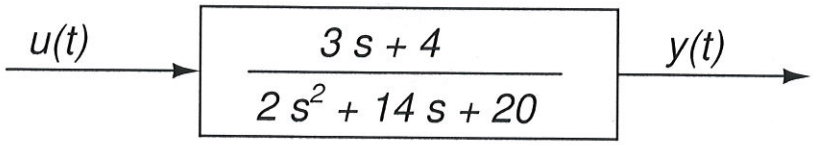



Figure 2: System element.

1. For the system element shown in figure 2, determine the governing differential equation.

$2\ddot{y}(t) + 14\dot{y}(t) + 20y(t) = 3\dot{u}(t) + 4u(t)$

2. Determine the impulse response for the system element given in figure 2. That is, find $y(t)$ when $u(t) = \delta(t)$.

$Y_1(s) = G_{uy}(s)\,U(s) = \dfrac{3s+4}{2s^2+14s+20} \cdot 1$

Find Poles

$s_{1,2} = \dfrac{-14 \pm \sqrt{14^2 - 4\cdot 2\cdot 20}}{2\cdot 2} = \dfrac{-14}{4} \pm \dfrac{\sqrt{196-160}}{4}$

$= -\dfrac{7}{2} \pm \dfrac{\sqrt{36}}{4} = -\dfrac{7}{2} \pm \dfrac{6}{4} = -\dfrac{4}{2}, -\dfrac{10}{2} = -2, -5$

Double check: $s=-2$, $8-28+20=0$ ✓, $s=-5$ $50-70+20=0$ ✓

Find Residues

$\dfrac{1\frac{1}{2}s+2}{s^2+7s+10} = \dfrac{C_1}{s+2} + \dfrac{C_2}{s+5} = \dfrac{C_1(s+5)+C_2(s+2)}{s^2+7s+10} = \dfrac{-1/3}{s+2} + \dfrac{11/6}{s+5}$

$(C_1+C_2) = 1\frac{1}{2}$ $\quad 3C_1 = -1$ $\quad C_2 = 1\frac{1}{2} - C_1$

$5C_1 + 2C_2 = 2$ $\quad C_1 = -1/3$ $\quad = \frac{9}{6} - \left(-\frac{2}{6}\right) = \frac{11}{6}$

$y(t) = -\frac{1}{3}e^{-2t} + \frac{11}{6}e^{-5t}$

**Problem 3.**

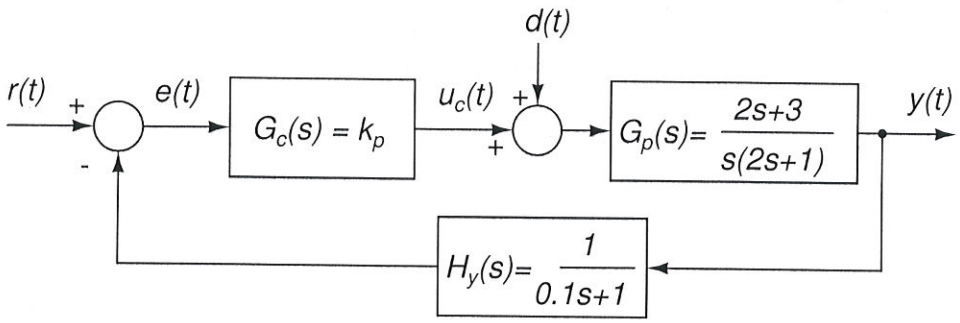



Figure 3: Closed-loop system.

$D_1 = s(2s+1)$
$D_2 = .1s+1$

1. For the system of figure 3, determine the transfer functions:

   (a) Response of the system to the reference input, $T_{ry}(s)$.

$$T_{ry}(s) = \frac{G_c G_p}{1 + G_c G_p H_y} = \frac{k_p \frac{2s+3}{D_1(s)}}{1 + k_p \frac{2s+3}{D_1(s)} \frac{1}{D_2(s)}} = \frac{k_p (2s+3) D_2(s)}{D_1(s) D_2(s) + k_p (2s+3)}$$

$$= \frac{k_p (2s+3)(.1s+1)}{s(2s+1)(.1s+1) + k_p (2s+3)}$$

| $s^3$ | $s^2$ | $s^1$ | $s^0$ |
|---|---|---|---|
| .2 | 2 | 1 | $3k_p$ |
| | .1 | $2k_p$ | |

   (b) Errors caused by the disturbance, $T_{de}(s)$.

$$T_{ry}(s) = \frac{k_p (.2s^2 + 2.3s + 3)}{.2s^3 + 2.1s^2 + (1+2k_p)s + 3k_p}$$

$$T_{de}(s) = \frac{G_p H_y (-1)}{1 + G_c G_p H_y} = \frac{(-1) \frac{2s+3}{D_1} \frac{1}{D_2}}{1 + k_p \frac{2+3}{D_1} \frac{1}{D_2}} = \frac{-1 (2s+3)}{D_1 D_2 + k_p (2+3)}$$

$$= \frac{-(2s+3)}{.2s^3 + 2.1s^2 + (1+2k_p)s + 3k_p}$$

## Problem 4.

Indicate whether each statement applies more substantially to a Laplace Transform, to a Transfer Function or to both (circle one).

1. It depends on system parameters. L.T. (T.F.) Both
2. It is determined from a time domain signal. (L.T.) T.F. Both
3. It has an inverse. (L.T.) T.F. Both
4. It has units. L.T. T.F. (Both)
5. The time-domain function can be found by partial fraction expansion. (L.T.) T.F. Both
6. It is a rational polynomial. L.T. T.F. (Both)

## Extra Credit

## Dynamic Equivalent

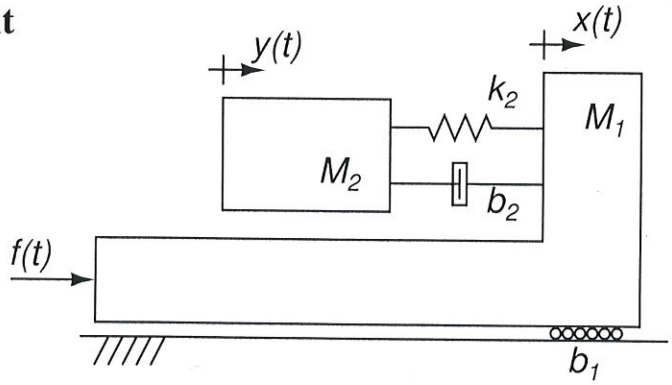



Figure 4: A mass ($M_1$) with passive vibration damper. $b_1$ models the friction between $M_1$ and the base.

In figure 4, a moving mass $M_1$ is connected to a passive damper, $M_2$. Parameters $M_2$, $k_2$ and $b_2$ can be tuned to reduce vibration over a range of frequencies. The John Hancock tower in Boston has a problem with building sway in high winds, and has a damping system on the $58^{th}$ floor similar to figure 4 (two masses $M_2$, each 300 tons, $k_2$ and $b_2$ provided by motor drive and feedback control).

Determine the electrical circuit that is dynamically equivalent to the mechanism of figure 4.

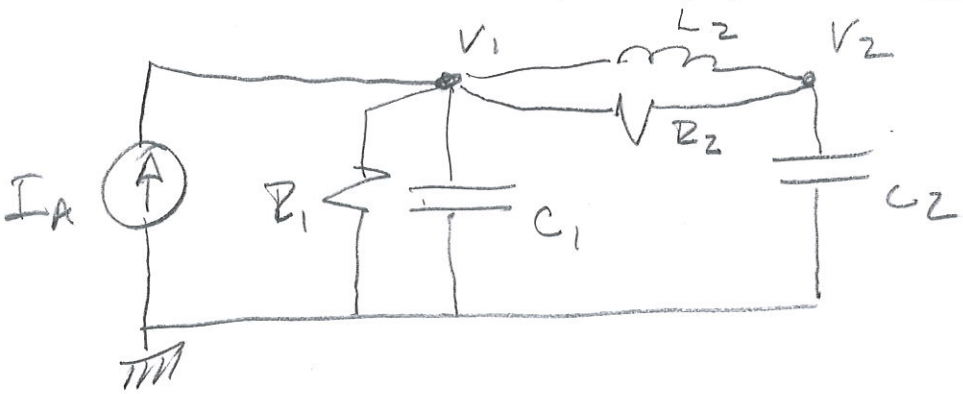
1. How many of each type of component will be included,

   (a) Resistors 2 (b) Capacitors 2 (c) Inductors 1

2. How many non-ground nodes will the circuit have ? 2

3. How will the external, applied force $f(t)$ be modeled in the circuit ? $I_A(t)$

4. Draw the schematic of the dynamically equivalent electrical circuit.




5. Indicate the component value of each component in your electrical circuit, in terms of the parameters of the mechanical system in figure 4.

$C_1 = M_1$ $L_2 = 1/k_2$ $R_1 = \frac{1}{b_1}$

$C_2 = M_2$

$R_2 = \frac{1}{b_2}$